Department of Electrical and Electronics Engineering Faculty Members

Birla Institute of Technology, Mesra, Jaipur Campus

Dr. Prashant Singh Rathore

Joined Institute on: 2-Mar-2006

Assistant Professor

Electrical and Electronics Engineering

PhD, M.Tech. (Power System)

CONTACT ADDRESS

27, MIA Malviya Nagar Jaipur

Phone Office: 0141 4019805

Phone Residence: 0141 4019805

Email: psrathore@bitmesra.ac.in

WORK EXPERIENCE

Teaching: 22 Years

PROFESSIONAL BACKGROUND

ACADEMIC QUALIFICATIONS

Ph.D. (2020)

In Solar PV system from Maharana Pratap University of Agriculture and Technology, Udaipur

M.E. (2003)

Master of Engineering in Power System Department of Electrical Engineering, from Malviya National Institute of Technology [Formerly known as Malviya Regional Engineering College] Jaipur, Rajasthan University, Rajasthan, (India).

B.E. (1997)

Bachelor of Engineering in Electronics & Power from Nagpur University, Nagpur, Maharashtra, (India)

PRESENT POSITIONS

1. Assistant Professor and In-charge of Electrical and Electronics Engineering Department at Birla Institute of Technology, Mesra, Ranchi, Jaipur Campus.

2. In-charge of Electronics & Communication Engineering Department at Birla Institute of Technology, Mesra, Ranchi, Jaipur Campus.

3. In-charge of Electrical Maintenance of the Birla Institute of Technology, Mesra, Ranchi, Jaipur Campus

PREVIOUS POSITIONS HELD

1. Technical Manager and In-charge of a NABL accredited (National Accreditation Board for Testing and Calibration Laboratories, New Delhi, India), Calibration Lab. Responsibilities: Calibration of different Electro-Technical Instruments such as Ammeter, Voltmeter (AC & DC), Ratiometer, Wattmeter, Digital

Multimeter, KV meter, Kelvin's Bridge, Micrometer, Vernier Calipers, CT, PT, Thermometer, Insulation test of cables and issuing NABL accredited calibration certificate to Industrial Instruments

2. Faculty Coordinator, Annual Sports event

3. Faculty Mentor, Annual Technical Fest

PHD SUPERVISION

1. Name of Scholar/Researcher: Mr. Deepak Chaturvedi

Roll No.: PhD/EE/10002/24

Guide/Co-Guide Status: Co-Guide

Date of Registration/Allotment: 26 July 2024

Date of Thesis Submission: Ongoing

Date of Award/Completion: Ongoing

2. Name of Scholar/Researcher: Mr. Manish Puroit

Roll No.: PhD/EC/10054/24

Guide/Co-Guide Status: Co-Guide

Date of Registration/Allotment: 2024

Date of Thesis Submission: Ongoing

Date of Award/Completion: (Not specified)

RESEARCH AREAS

1. Renewable Energy resources

2. Smart Grid

AWARD AND HONOURS

IEEE Senior Member, Institute of Electrical and Electronics Engineers (IEEE), USA (2025).

Elevated to the prestigious IEEE Senior Member grade in recognition of significant professional achievements, research contributions, and experience in electrical engineering and allied fields.

PUBLICATIONS

1. Deepak Verma, Rathore, P. S., Sharma, P., & Sharma, J. (2026, February 19–20). A deep learning–based virtual sensor for junction temperature estimation in IGBT/SiC inverter legs toward digital twin–enabled health monitoring. Paper presented at the 3rd International Conference on Emerging Technologies and Applications in Electrical Engineering (ETAEE-2026), National Institute of Technology Raipur, India (Online).

2. Singh, G., Chaturvedi, D., Deepak Verma, & Rathore, (2025). Recent advances in battery management system topologies for electric vehicles. In Proceedings of the 12th National Power Electronics Conference

(NPEC 2025) (pp. 1–6). National Institute of Technology Calicut, India. https://doi.org/10.1109/NPEC66512.2025.11450232

3. Kasinath Jena, Dhananjay Kumar, Ashutosh Kumar Singh, Prashant Singh Rathore & Deepak Verma (2023) "Single source multi gain 7-level switched-capacitor inverter topology with reduce device count" International Journal of Electronics Letters, DOI: 10.1080/21681724.2023.2267206 [SCI and SCOPUS indexed]

4. P. Gupta, R. Verma, Deepak Verma and P. S. Rathore, "Building a Stronger Grid: How Blockchain Can Improve Smart Grid Resilience," 2023 IEEE Renewable Energy and Sustainable E-Mobility Conference (RESEM), Bhopal, India, 2023, pp. 1-7, doi: 10.1109/RESEM57584.2023.10236304.

5. Y. Kumawat, S. Shukla, Deepak Verma and P. S. Rathore, "Wireless Energy Harvesting and Transfer: A Comprehensive Review of Recent Developments," 2023 IEEE Renewable Energy and Sustainable E-Mobility Conference (RESEM), Bhopal, India, 2023, pp. 1-4, doi: 10.1109/RESEM57584.2023.10236286.

6. K. Jain, Deepak Verma and P. S. Rathore, "Assessment of Blockchain based Energy Trading methods for Electric Vehicles," 2022 IEEE 6th International Conference on Condition Assessment Techniques in Electrical Systems (CATCON), Durgapur, India, 2022, pp. 6-11, doi: 10.1109/CATCON56237.2022.10077682.

7. R. Jangir, Deepak Verma and P. S. Rathore, "Implementation and Analysis of Inductor-less MPPT controlled Reflex Wave Charging of Lead-Acid Battery," 2022 IEEE 6th International Conference on Condition Assessment Techniques in Electrical Systems (CATCON), Durgapur, India, 2022, pp. 104-109, doi: 10.1109/CATCON56237.2022.10077625.

MEMBER OF PROFESSIONAL BODIES

Member IEEE

CURRENT SPONSORED PROJECTS

Research project entitled "Development of a Microcontroller Based Technology of a pressure sensitive bed for immovable Patients", Sponsored by Indian Council of Medical Research (ICMR), New Delhi, India.

Sanctioned amount for Research work: INR 9,65,941/-

Duration: (Sep. 2009 to Aug. 2011)

Dr. Abhishek Thakur

Joined Institute on: 12-Mar-2026

Assistant Professor

Electrical and Electronics Engineering

M.Tech, Ph.D.

CONTACT ADDRESS

Birla Institute of Technology (BIT) Jaipur Campus, Malviya Industrial Area, near Apex Circle, Jaipur, Rajasthan-302017

Narendra Sadan, Laheriasarai, Darbhanga, Bihar-846001

Phone Office: 8917443457

Phone Residence: 8917443457

Email: abhishekthakur9396@bitmesra.ac.in

WORK EXPERIENCE

Research: 4 Years

PUBLICATIONS

JOURNALS

1. Thakur, A., & Mishra, S. K. (2024). "An in-depth evaluation of deep learning-enabled adaptive approaches for detecting obstacles using sensor-fused data in autonomous vehicles." Engineering Applications of Artificial Intelligence, Elsevier, Vol. 133, Part F, Article No. 108550. ISSN: 0952-1976; Online ISSN: 1873-6769. DOI: https://doi.org/10.1016/j.engappai.2024.108550. Impact Factor: 8.0; CiteScore: 9.5; Indexed in Scopus and Science Citation Index Expanded (SCIE).

2. Thakur, A., & Mishra, S. K. (2025). "A comprehensive review on deep learning based adaptive methods for obstacle detection in autonomous ground vehicles using sensor fusion." International Journal of Heavy Vehicle Systems, Inderscience Publishers, DOI: https://doi.org/10.1504/IJHVS.2025.10072587. ISSN Online: 1741-5152; ISSN Print: 1744-232X. Impact Factor: 0.6; 5-Year Impact Factor: 0.8; CiteScore: 1.2. Indexed in Scopus, Science Citation Index Expanded (SCIE), Journal Citation Reports, Compendex/Ei, and J-Gate.

3. Thakur, A., Kumar, A., Mishra, S. K., Behera, S. K., Sethi, J., Sahu, S. S., & Swain, S. K. (2024). "Product Length Predictions with Machine Learning: An Integrated Approach Using Extreme Gradient Boosting." SN Computer Science, Springer Nature Singapore, Vol. 5, Article No. 659. DOI: https://doi.org/10.1007/s42979-024-02999-8. Electronic ISSN: 2661-8907; Print ISSN: 2662-995X; CiteScore: 5.8; SJR: 0.565; SNIP: 1.137; Indexed in Scopus and other major academic databases.

4. Thakur, A., Das, S., Mishra, S. K., & Swain, S. K. (2025). "Adaptive stochastic gradient descent with least angle regression enhanced navigation: intelligent path planning in cluttered environments for autonomous robots." Transactions on Energy Systems and Engineering Applications, Universidad Tecnológica de Bolívar, Vol. 6, No. 2, pp. 1–26. DOI: https://doi.org/10.32397/tesea.vol6.n2.602. ISSN: 2745-0120; Indexed in Scopus, DOAJ, Dialnet, and Google Scholar.

5. Thakur, A., Mishra, S. K., Bhutani, M., Dash, P. P., Upadhyay, P. K., & Sahu, S. S. (2026). "Deep learning approaches for detecting malignant melanoma in dermoscopic imagery." International Journal of Information Technology, Springer Nature, Vol. 18, pp. 91–109. DOI: https://doi.org/10.1007/s41870-025-02651-5. Electronic ISSN: 2511-2112; Print ISSN: 2511-2104; Indexed in Scopus.

6. Das, S., Kumar, A., Thakur, A., & Mishra, S. K. (2026). A noise resilient wavelet packet transform assisted CNN-GRU framework for remaining useful life prediction of rolling bearings. Engineering Research Express, 8(13), 135411 (ESCI). https://doi.org/10.1088/2631-8695/ae841c

BOOK CHAPTERS

1. Thakur, A., & Mishra, S. K. (2023). "Review on Vision-Based Control Using Artificial Intelligence in Autonomous Ground Vehicle." In R. P. Yadav, S. J. Nanda, P. S. Rana, & M.-H. Lim (Eds.), Proceedings of the International Conference on Paradigms of Computing, Communication and Data Sciences, Algorithms for Intelligent Systems, Springer Nature Singapore, pp. 617–626. DOI: https://doi.org/10.1007/978-981-19-8742-7_49. Print ISBN: 978-981-19-8741-0; Online ISBN: 978-981-19-8742-7; First Online: 24 February 2023.

2. Das, S., Kumari, R., Kumar, A., Thakur, A., & Singh, R. K. (2024). "Unveiling Superior Lane Detection Techniques Through the Synergistic Fusion of Attention-Based Vision Transformers and Dense Convolutional Neural Networks." In A. E. Hassanien, S. Anand, A. Jaiswal, & P. Kumar (Eds.), Innovative Computing and Communications, ICICC 2024, Lecture Notes in Networks and Systems, Vol. 1020, Springer Nature Singapore, pp. 15–27. DOI: https://doi.org/10.1007/978-981-97-3588-4_2. Print ISBN: 978-981-97-3587-7; Online ISBN: 978-981-97-3588-4; First Online: 24 July 2024.

3. Kumar, A., Thakur, A., Mishra, S. K., & Swain, S. K. (2024). "LDCCAES: A Concomitant Perception Methodology Facilitating Real-Time Detection and Estimation of Median-Lane Positioning for Prototype Autonomous Vehicle." In S. J. Nanda, R. P. Yadav, A. H. Gandomi, & M. Saraswat (Eds.), Data Science and Applications, ICDSA 2023, Lecture Notes in Networks and Systems, Vol. 821, Springer Nature Singapore, pp. 169–179. DOI: https://doi.org/10.1007/978-981-99-7814-4_14. Print ISBN: 978-981-99-7813-7; Online ISBN: 978-981-99-7814-4; First Online: 25 February 2024.

4. Thakur, A., Kumar, A., & Mishra, S. K. (2024). "Control Techniques for Vision-Based Autonomous Vehicles for Agricultural Applications: A Meta-analytic Review." In H. Sharma, A. Chakravorty, S. Hussain, & R. Kumari (Eds.), Artificial Intelligence: Theory and Applications, AITA 2023, Lecture Notes in Networks and Systems, Vol. 843, Springer Nature Singapore, pp. 1–14. DOI: https://doi.org/10.1007/978-981-99-8476-3_1. Print ISBN: 978-981-99-8475-6; Online ISBN: 978-981-99-8476-3.

5. Das, S., Thakur, A., Mishra, S. K., Swain, S. K., & Kumari, R. (2024). "Development of Indoor Autonomous Mobile BOT for Static Obstacle Avoidance." In S. J. Nanda, R. P. Yadav, A. H. Gandomi, & M. Saraswat (Eds.), Data Science and Applications, ICDSA 2023, Lecture Notes in Networks and Systems, Vol. 820, Springer Nature Singapore, pp. 43–55. DOI: https://doi.org/10.1007/978-981-99-7817-5_5. Print ISBN: 978-981-99-7816-8; Online ISBN: 978-981-99-7817-5.

6. Lohani, M. P., Raj, A., Rawat, P., Thakur, A., & Mishra, S. K. (2026). "Deep Q-Learning for Autonomous Path Following: A Multi-environment Study with Obstacle Avoidance." In H. Sharma, A. Bhatt, C. Modi, & A. Engelbrecht (Eds.), Computer Vision and Robotics, CVR 2025, Lecture Notes in Networks and Systems, Vol. 1770, Springer, Cham, pp. 453–466. DOI: https://doi.org/10.1007/978-3-032-14038-8_36. Print ISBN: 978-3-032-14037-1; Online ISBN: 978-3-032-14038-8.

7. Kumari, R., Das, S., Thakur, A., Kumar, A., & Singh, R. K. (2024). "Autonomous Vehicles." In K. Malik, M. Sharma, S. Deswal, U. Gupta, D. Agarwal, & Y. O. B. Al Shamsi (Eds.), Explainable Artificial Intelligence for

Autonomous Vehicles: Concepts, Challenges, and Applications, CRC Press/Taylor & Francis, Chapter 1, pp. 1–24. DOI: https://doi.org/10.1201/9781003502432-1. eBook ISBN: 9781003502432.

8. Das, S., Kumari, R., Thakur, A., Singh, R. K., & Nigam, A. (2024). "Path Planning for Autonomous Ground Vehicles by Applying Modified Harris Hawks Optimization Technique." In M. Pant, K. Deep, & A. Nagar (Eds.), Proceedings of the 12th International Conference on Soft Computing for Problem Solving, SocProS 2023, Lecture Notes in Networks and Systems, Vol. 994, Springer Nature Singapore, pp. 161–171. DOI: https://doi.org/10.1007/978-981-97-3180-0_11. Print ISBN: 978-981-97-3179-4; Online ISBN: 978-981-97-3180-0; First Online: 23 July 2024.

CONFERENCES

1. Thakur, A., Gupta, V., Joshi, A., Mishra, S. K., & Swain, S. K. (2025). "α-Blend MPC: A Curvature-Adaptive Dual-Weight Model Predictive Control for Autonomous Vehicle Navigation." In Proceedings of the 2025 IEEE 4th International Conference on Smart Technologies for Power, Energy and Control (STPEC 2025), IEEE, NIT Goa, India, 10–13 December 2025. IEEE Xplore Document No.: 10826927; Conference Paper.

2. Thakur, A., Gupta, V., Joshi, A., Mishra, S. K., & Swain, S. K. (2025). "α-Blend MPC: A Curvature-Adaptive Dual-Weight Model Predictive Control for Autonomous Vehicle Navigation." In Proceedings of the 2025 IEEE 4th International Conference on Smart Technologies for Power, Energy and Control (STPEC 2025), IEEE, NIT Goa, India. DOI: https://doi.org/10.1109/STPEC66316.2025.11490864. IEEE Xplore Document No.: 11490864; Conference Paper.

INTELLECTUAL PROPERTY RIGHTS (IPR)

Ankit Kumar, Abhishek Thakur, Sudhansu Kumar Mishra, Subrat Kumar Swain, and Sitanshu Sekhar Sahu, "An Autonomous Vehicle Navigation System and Method Thereof," Indian Patent / Indian Design Patent, Application No. 202331066870, Indian Patent Office, Government of India, filed on 05 October 2023 and published in Patent Journal on 03 November 2023.

Dr. Deepak Verma

Joined Institute on: 9-Jun-2017

Assistant Professor

Electrical and Electronics Engineering

Ph.D

CONTACT ADDRESS

Kh. No. 101/1 Ramnagaria, Jagatpura, Jaipur

Betul, Madhya Pradesh

Phone Office: 9893041416

Phone Residence: 9893853816

Email: deepakverma@bitmesra.ac.in

WORK EXPERIENCE

Teaching: 15 Years

PROFESSIONAL BACKGROUND

PAST EXPERIENCE

Position: Assistant Professor

Organization: Maulana Azad National Institute of Technology Bhopal (NIT Bhopal)

Duration: 06 Years, 2011-17

ADMINISTRATIVE EXPERIENCE AT NIT BHOPAL

Member of BOS (the Board of Studies of the Institute).

Member of Institute committee for Physical Stock Verification.

Member of DUPC (Deptt. Undergraduate Program Committee).

Member of committee constituted for NBA Accreditation of PG programs M.Tech. (PS) and M.Tech. (ED).

Member of core committee constituted for NIRF information.

EDUCATION QUALIFICATIONS

1. Exam/Degree: Matriculation

Specialization: --

University: Board of Secondary Education Bhopal

Year: 2002

2. Exam/Degree: Higher Secondary

Specialization: Physics Chemistry Mathematics

University: Board of Secondary Education Bhopal

Year: 2004

3. Exam/Degree: UG: B.E.

Specialization: Electrical & Electronics Engg.

University: Rajiv Gandhi Technical University Bhopal

Year: 2008

4. Exam/Degree: PG: M.Tech.

Specialization: Power System

University: NIT Bhopal

Year: 2010

5. Exam/Degree: Ph.D.

Specialization: Electrical Engineering

University: NIT Bhopal

Year: 2016

INSTITUTE-LEVEL RESPONSIBILITIES

1. Chairperson, Institute Industry Cell, BIT Jaipur Campus (03 December 2025 – Present)

Responsible for strengthening institute–industry interaction, promoting industry collaborations, internships, expert talks, and collaborative projects.

2. Coordinator, BIT Jaipur Incubation Center, BIT Jaipur Campus (06 November 2025 – Present)

Coordinating the establishment and operational planning of the incubation center to promote innovation, entrepreneurship, startup development, and industry engagement.

3. Assistant, Faculty In-Charge Electrical Maintenance, BIT Jaipur Campus (07 July 2020 – Present)

Responsible for planning, supervision, and execution of campus electrical maintenance activities; contributed to enhancement of institute electrical load capacity from 150 kVA to 500 kVA.

4. Member, Entrepreneurship Cell, BIT Jaipur Campus (19 October 2022 – Present)

Contributing to entrepreneurship awareness, startup promotion, innovation activities, and student engagement programs.

5. Member, GATE Advisory Cell, BIT Jaipur Campus (23 August 2024 – Present)

Guiding and motivating students for GATE preparation, higher education opportunities, and competitive examinations.

6. Member, Institute Innovation Council (IIC), BIT Jaipur Campus (2025 – Present)

Supporting innovation-driven activities, startup initiatives, intellectual property awareness, and entrepreneurship development.

7. Member, Institute Innovation Cell, BIT Jaipur Campus (2022 – 2023)

Assisted in organizing innovation-related programs and promoting a culture of creativity and problem-solving among students.

8. Member, Institution Industry Cell, BIT Jaipur Campus (2022 – 2023)

Supported industry interaction activities and facilitated academic–industry engagement initiatives.

9. Member, Intellectual Property Rights (IPR) Cell, BIT Jaipur Campus (2022 – 2023)

Contributed to activities related to intellectual property awareness, patent filing, and innovation promotion.

10. Coordinator, Microsoft Teams Platform, BIT Jaipur Campus (05 August 2020 – Present)

Managed and coordinated institute-wide deployment and operation of the Microsoft Teams platform for academic and administrative activities.

11. Coordinator, Annual Tech Fest – TECHVIBES 2023, BIT Jaipur Campus (Calendar Year 2023)

Coordinated planning and execution of institute-level technical events, competitions, and student innovation activities.

12. Coordinator, Annual Tech Fest – TECHVIBES 2024, BIT Jaipur Campus (Calendar Year 2024)

Led the organization of technical events, project exhibitions, and student participation activities under the annual techno-management festival.

13. Organising Secretary, International Conference on Innovations in Computing and Applications (ICICA-24), BIT Jaipur Campus (2024)

Coordinated conference planning, technical sessions, publication activities, and overall event management for the SERB/ANRF-supported international conference.

14. SPOC, Smart India Hackathon (SIH) 2019, BIT Jaipur Campus (2019)

Coordinated student participation, mentoring, and institute-level activities related to Smart India Hackathon.

15. Event Co-Coordinator, Smart India Hackathon (SIH) 2020, BIT Jaipur Campus (31 January 2020)

Assisted in organizing and coordinating hackathon-related activities hosted at the institute.

DEPARTMENTAL-LEVEL RESPONSIBILITIES

1. UG Registration Coordinator, Department of Electrical & Electronics Engineering, BIT Jaipur Campus (2017 – Present)

Coordinating undergraduate student registration, course enrollment verification, academic record maintenance, and related departmental academic activities.

2. Laboratory In-Charge, Electrical Machines Laboratory-I, Electrical Machines Laboratory-II and Power System Laboratory, Department of Electrical & Electronics Engineering, BIT Jaipur Campus (2017 – Present)

Responsible for laboratory planning, equipment maintenance, practical course delivery, safety compliance, laboratory modernization, and outcome-based practical learning activities.

3. Mentor, Technical Club – Power Electronics & Devices, Department of Electrical & Electronics Engineering, BIT Jaipur Campus (2018 – Present)

Mentoring students in hardware-based projects, technical competitions, project exhibitions, and innovation-oriented activities related to power electronics and electrical engineering.

4. Faculty Coordinator for Industrial Visits, Summer Training and Internship Activities, Department of Electrical & Electronics Engineering, BIT Jaipur Campus (2018 – Present)

Coordinating industrial visits, internships, summer training programs, industry interaction activities, and experiential learning opportunities to enhance students' practical exposure and employability.

PH.D. SUPERVISION

1. Name of Scholar/Researcher: Mr. Deepak Chaturvedi

Roll No.: PhD/EE/10002/24

Guide/Co-Guide Status: Guide

Date of Registration/Allotment: 26 July 2024

Date of Thesis Submission: Ongoing

Date of Award/Completion: Ongoing

EVENTS / CONFERENCES / WORKSHOPS / FDPs ORGANIZED

1. Organising Secretary, International Conference on Innovations in Computing and Applications (ICICA-24), BIT Jaipur Campus, 2024.

2. Organising Secretary, Organized Short Term Training Programme (STTP) on "Introduction to MATLAB & SIMULINK: A Hands-On Training Programme", 24–28 May 2020.

3. Organized Webinar on "Getting Started on Final Year Projects" for pre-final year students, 29 April 2020.

4. Organized Webinar on "Review Based or Software Based Project Preparation" for final year students, 05 May 2020.

5. Organized Seminar on "GATE Preparation" by alumnus Mr. Lakshya Gupta, 07 February 2020.

6. Organized Seminar on "Career Prospects after Electrical Engineering" by Mr. Rajesh Goyal, Director, Kanak Automation, 05 February 2020.

CONFERENCE/SEMINAR/CONCLAVE DETAILS

1. Conference Title: Science and Engineering Research Board (SERB) Sponsored International Conference on Innovations in Computing and Applications

Venue: Birla Institute of Technology Off Campus Jaipur

Organised by: Birla Institute of Technology Off Campus Jaipur

Date: May 22, 2024 To May 23, 2024

Category: International

Role: Organising Secretary

2. Conference Title: IEEE International Students' Conference on Electrical, Electronics & Computer Science 2018 (SCEECS 2018)

Venue: Bhopal

Organised by: Maulana Azad National Institute of Technology Bhopal.

Date: Feb 24, 2018 To Feb 25, 2018

Category: International

Role: Member of organizing committee

3. Conference Title: IEEE International Students' Conference on Electrical, Electronics & Computer Science 2016 (SCEECS 2016)

Venue: Maulana Azad National Institute of Technology, Bhopal

Organised by: Maulana Azad National Institute of Technology, Bhopal

Date: Mar 05, 2016 To Mar 06, 2016

Category: International

Role: Member of organizing committee

4. Conference Title: International Conference on Recent Trends in Engineering and Sciences (RTES 2023)

Venue: Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat

Organised by: Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat

Date: May 02, 2023 To May 03, 2023

Category: International

Role: Presenter

5. Conference Title: International Conference on "Electrical Power and Energy Systems" (ICEPES-2024)

Venue: Maulana Azad National Institute of Technology, Bhopal

Organised by: Maulana Azad National Institute of Technology, Bhopal

Date: Jun 21, 2024 To Jun 22, 2024

Category: International

Role: Session Chair

6. Conference Title: National Conference on "Advances in Mechanical, Electronics and Electrical Engineering, AMEEE-2021"

Venue: Maulana Azad National Institute of Technology, Bhopal

Organised by: Maulana Azad National Institute of Technology, Bhopal

Date: Apr 15, 2021 To Apr 16, 2021

Category: National

Role: Session Chair

7. Conference Title: 2nd International Conference on Electrical Power and Energy Systems, ICEPES-2021

Venue: Maulana Azad National Institute of Technology, Bhopal

Organised by: Maulana Azad National Institute of Technology, Bhopal

Date: Dec 10, 2021 To Dec 11, 2021

Category: International

Role: Session Chair

8. Conference Title: IEEE International Symposium on Nanoelectronic and Information Systems (IEEE-iNIS 2017)

Venue: Oriental Institute of Science & Technology, Bhopal.

Organised by: Oriental Institute of Science & Technology, Bhopal.

Date: Dec 18, 2017 To Dec 20, 2017

Category: International

Role: Member of organizing committee

9. Conference Title: IEEE International Conference on Electrical Power and Energy Systems (ICEPES-2016)

Venue: Maulana Azad National Institute of Technology, Bhopal

Organised by: Maulana Azad National Institute of Technology, Bhopal

Date: Dec 14, 2016 To Dec 16, 2016

Category: International

Role: Member of organizing committee

10. Conference Title: 5-day Face-to-Face UHV-II FDP organized by All India Council for Technical Education (AICTE)

Venue: Sagar Institute of Research and Technology, Indore

Organised by: All India Council for Technical Education (AICTE)

Date: Aug 02, 2022 To Aug 06, 2022

Category: National

Role: Attendee

11. Conference Title: Delivered expert lecture in 10 days training program, sponsored by MNRE and NISE on Rooftop Solar Grid Engineer course organized by Rajasthan Electronics & Instruments Ltd. (REIL), Jaipur on "Overview and selection criteria for Solar PV Modules 06th March 2019 to 19th March 2019."

Venue: Rajasthan Electronics & Instruments Ltd. (REIL), Jaipur

Organised by: Rajasthan Electronics & Instruments Ltd. (REIL), Jaipur

Date: Mar 06, 2019 To Mar 19, 2019

Category: National

Role: Speaker

RESEARCH AREAS

RESEARCH INTEREST

Performance optimization of Solar PV Array

Maximum Power Point Tracking

DC-DC Converter Design

Multilevel Inverter

Grid Integration of Solar PV Array

AWARD AND HONOURS

1. IEEE Senior Member, Institute of Electrical and Electronics Engineers (IEEE), USA (2025).

Elevated to the prestigious IEEE Senior Member grade in recognition of significant professional achievements, research contributions, and experience in electrical engineering and allied fields.

2. Recognized in the World's Top Scientists List, AD Scientific Index (2022–2025).

Recognized based on research publications, citations, and academic impact in Electrical Engineering, Renewable Energy Systems, and Power Electronics.

3. Achieved 1000+ Google Scholar Citations (2025).

Crossed the milestone of 1000 citations on Google Scholar, reflecting sustained research impact and scholarly contribution.

4. MHRD Scholarship Recipient during M.Tech. Programme at MANIT Bhopal.

Awarded Government of India scholarship based on academic merit and GATE qualification.

5. GATE Qualified (Electrical Engineering) with All India Rank and eligibility for higher studies and MHRD fellowship.

6. NCC 'A' Certificate Holder, National Cadet Corps (NCC), Government of India.

PUBLICATIONS

INTERNATIONAL JOURNALS

1. P. Sharma, S. Baloda, Deepak Verma, V. Janyani, R. Sharma and N. Gupta, "Flexible Resistive Strain Sensor based on PDMS Coated MWCNTs and MWCNTs/GNP for Wrist Pulse and Respiration Monitoring," in IEEE Journal on Flexible Electronics, doi: 10.1109/JFLEX.2025.3591416. [SCOPUS: https://www.scopus.com/sourceid/21101266872, Journal Quartile: Q2, CiteScore 2024: 3.4]

2. P. Sharma, S. Baloda, Deepak Verma, V. Janyani, R. Sharma and N. Gupta, "Multi-Wall Carbon Nanotube/ Polydimethylsiloxane composites based dry electrodes for bio-signal detection," in IEEE Journal on Flexible Electronics, doi: 10.1109/JFLEX.2024.3367648. [SCOPUS: https://www.scopus.com/sourceid/21101266872, Journal Quartile: Q2, CiteScore 2024: 3.4]

3. P. Sharma, R. Sharma, V. Janyani, Deepak Verma, "Development of a multi-modal graphene nanoparticles (GNP)- Polydimethylsiloxane (PDMS) flexible sensor for human activity monitoring and health assessment" International Journal of Electrochemical Science, Volume 18, Issue 9, 2023, 100236, https://doi.org/10.1016/j.ijoes.2023.100236. [Web of Science (WoS) – SCIE: https://wos-journal.info/journalid/20590 Journal Quartile: Q3, Impact Factor 2.4]

4. Kasinath Jena, Dhananjay Kumar, Ashutosh Kumar Singh, Prashant Singh Rathore & Deepak Verma (2023) "Single source multi gain 7-level switched-capacitor inverter topology with reduce device count" International Journal of Electronics Letters, DOI: 10.1080/21681724.2023.2267206 [SCOPUS: https://www.scopus.com/sourceid/21100854223, Journal Quartile: Q3, CiteScore 2024: 1.9]

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MEMBER OF PROFESSIONAL BODIES

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CURRENT SPONSORED PROJECTS

SERB Sponsored International Conference Grant (Seminar/Symposia)

Project/Activity Title: International Conference on Innovations in Computing and Applications (ICICA-24)

Funding Agency: Science and Engineering Research Board (SERB), Government of India (Now ANRF)

Role: Principal Investigator / Organising Secretary

File No.: SSY/2024/000409

Year: 2024

Amount Sanctioned: ₹1,50,000

Status: Completed

Outcome: Successfully organized the SERB-sponsored International Conference on Innovations in Computing and Applications (ICICA-24), facilitating dissemination of research, academic networking, and publication of conference proceedings in AIP Conference Proceedings.

https://printorders.aip.org/proceedings/3253

TEXT AND REFERENCE BOOKS

A. BOOK CHAPTER AND CONFERENCE PAPER

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Publisher Name: Springer, Singapore

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B. BOOK CHAPTER

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Publisher Name: Nova Science Publishers, Inc., NY, USA

ISBN: 978-1-53618-164-7

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INTERNATIONAL PATENT

1. Patent Title: An Efficient Power Distribution Based on IoT-Fog Resource for Effective Proxy Negotiation for Traffic Reduction

Patent No.: 2020104365

Award/Grant Date: 03/03/2021

Agency/Country: IP Australia, Australian Government

Status: Granted

NATIONAL PATENT

1. Patent Title: System on Chip Controller for Battery Charging for Electric Vehicle Applications

Application No.: 202211028719

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Agency/Country: Intellectual Property India, Government of India

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2. Patent Title: Flexible Sensor System for Protecting Battery Deformation and Related Methods

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Agency/Country: Intellectual Property India, Government of India

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REGISTERED INDIAN DESIGN

1. Design Title: Solar Powered Smart Helmet for Traffic Police

Design Registration No.: 423455-001

Registration Date: 17/07/2024

Agency/Country: Intellectual Property India, Government of India

Status: Registered Design

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QUALIFICATION

M.Tech, Ph.D (Thesis submitted)

AREA OF INTEREST

State and parameter estimation, nonlinear systems, data-driven estimation, Control systems, Kalman filters

FIELD OF INTEREST

Data driven Kalman filters, EV Battery state estimation, Reinforcement Learning, Deep learning